UAV Flight Analysis Report

Efficient Flight Profile

Report Generated: 2026-03-21 20:52:49

Flight Profile: Efficient

Data Source: flight_efficient_2026-03-21_20-40-14.csv

Flight Duration: 12.1 minutes

Executive Summary

This report presents a comprehensive analysis of an autonomous UAV inspection flight covering 1369 meters over 12.1 minutes. The mission utilized a Efficient flight profile with a maximum speed of 6.2 m/s, completing 24 waypoints while consuming 45% battery capacity. The flight maintained an average altitude of 14.3m with high precision positioning throughout the mission.

Key Performance Indicators

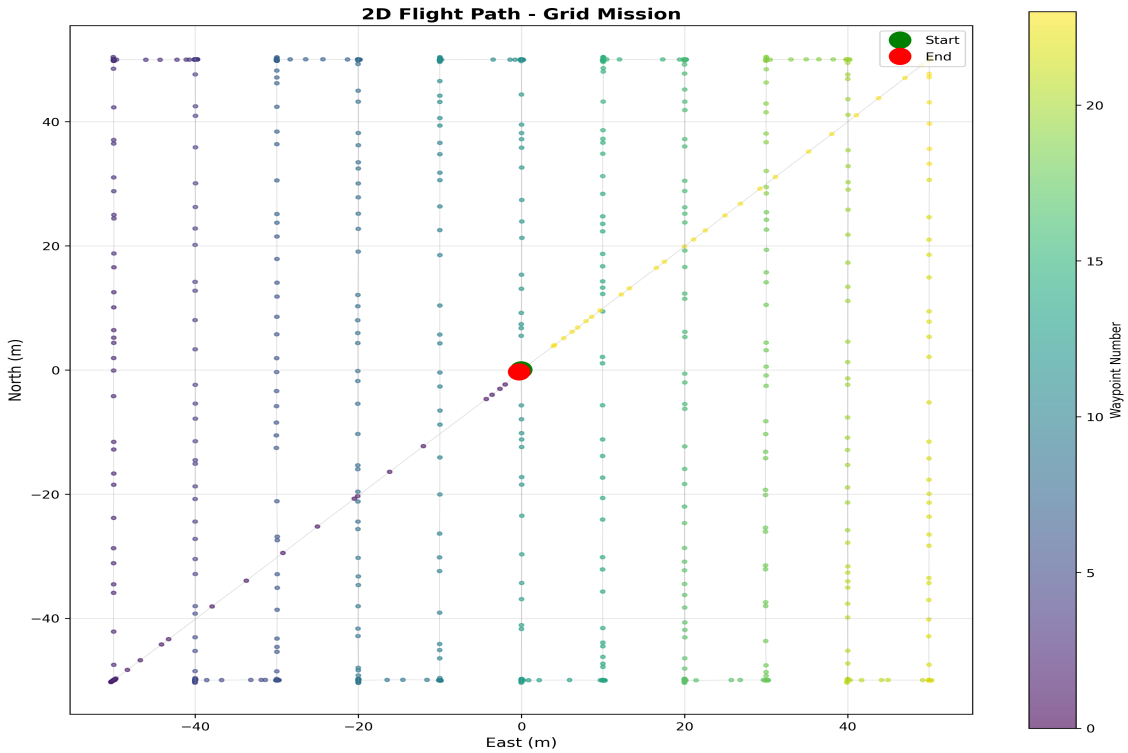
Metric	Value	Rating
Mission Completion	100%	✓ Excellent
Battery Efficiency	3.72%/min	✓ Good
Flight Stability	147 events	■ Moderate
Altitude Precision	±0.86m	✓ Excellent

Detailed Flight Statistics

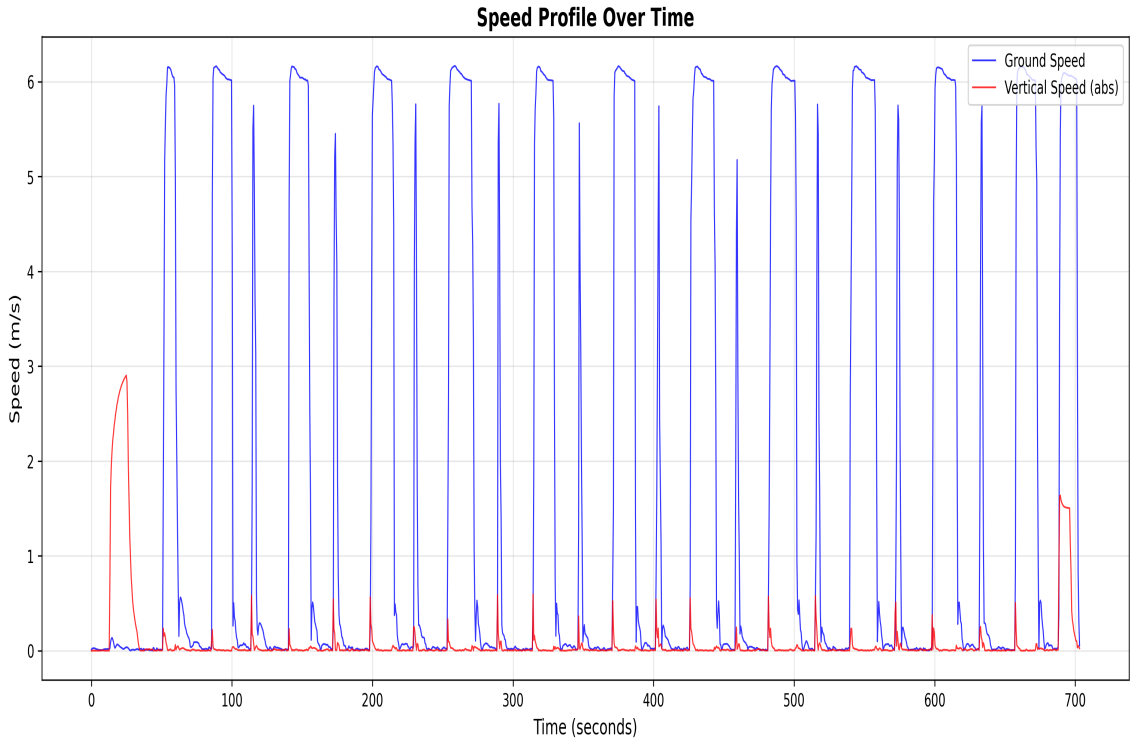
Category	Metric	Value	Unit
Mission	Total Duration	12.1	minutes
	Total Distance	1368.7	meters
	Waypoints	24	points
Speed	Maximum	6.17	m/s
	Average	1.91	m/s
	Median	0.08	m/s
Altitude	Maximum	15.20	meters
	Average	14.34	meters
	Minimum	-0.07	meters
Battery	Starting Level	96	%
	Ending Level	51	%
	Total Used	45	%
	Consumption Rate	3.72	%/min
Performance	High Accel Events	147	events
	Peak Acceleration	10.55	m/s^2

Flight Visualizations

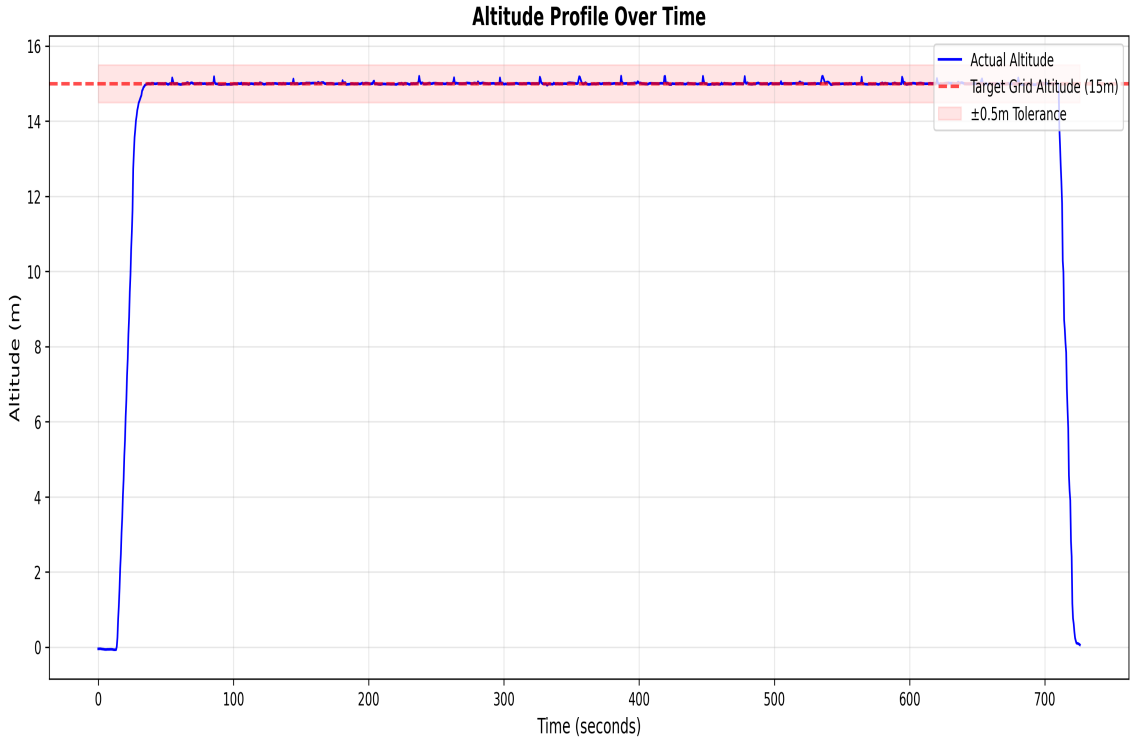
2D Flight Path (Top-Down View)



Speed Profile Over Time



Altitude Profile



Recommendations

- Mission uses significant battery - plan for recharging between flights
- Estimated 1.1 missions possible per battery (50% usable capacity)